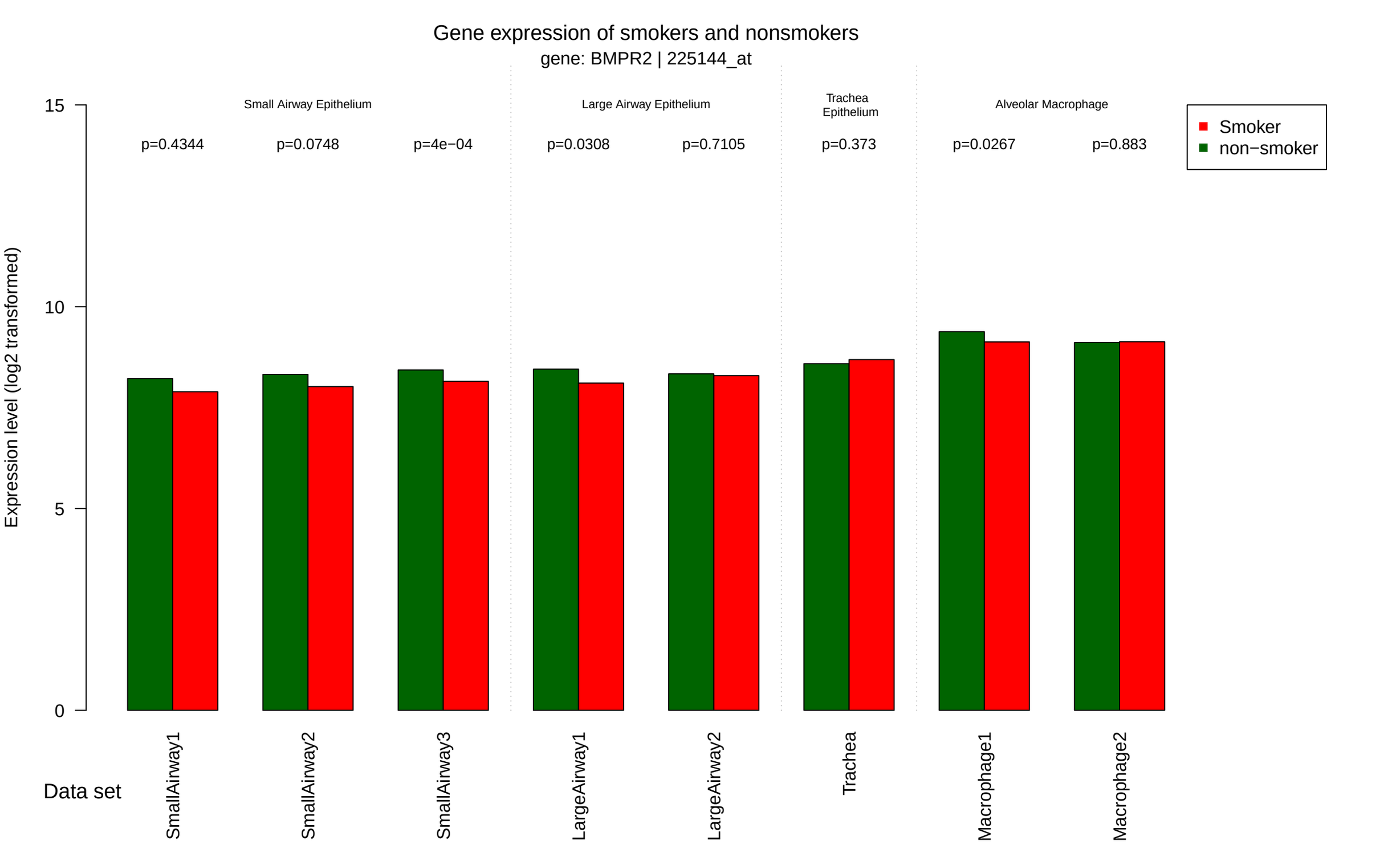
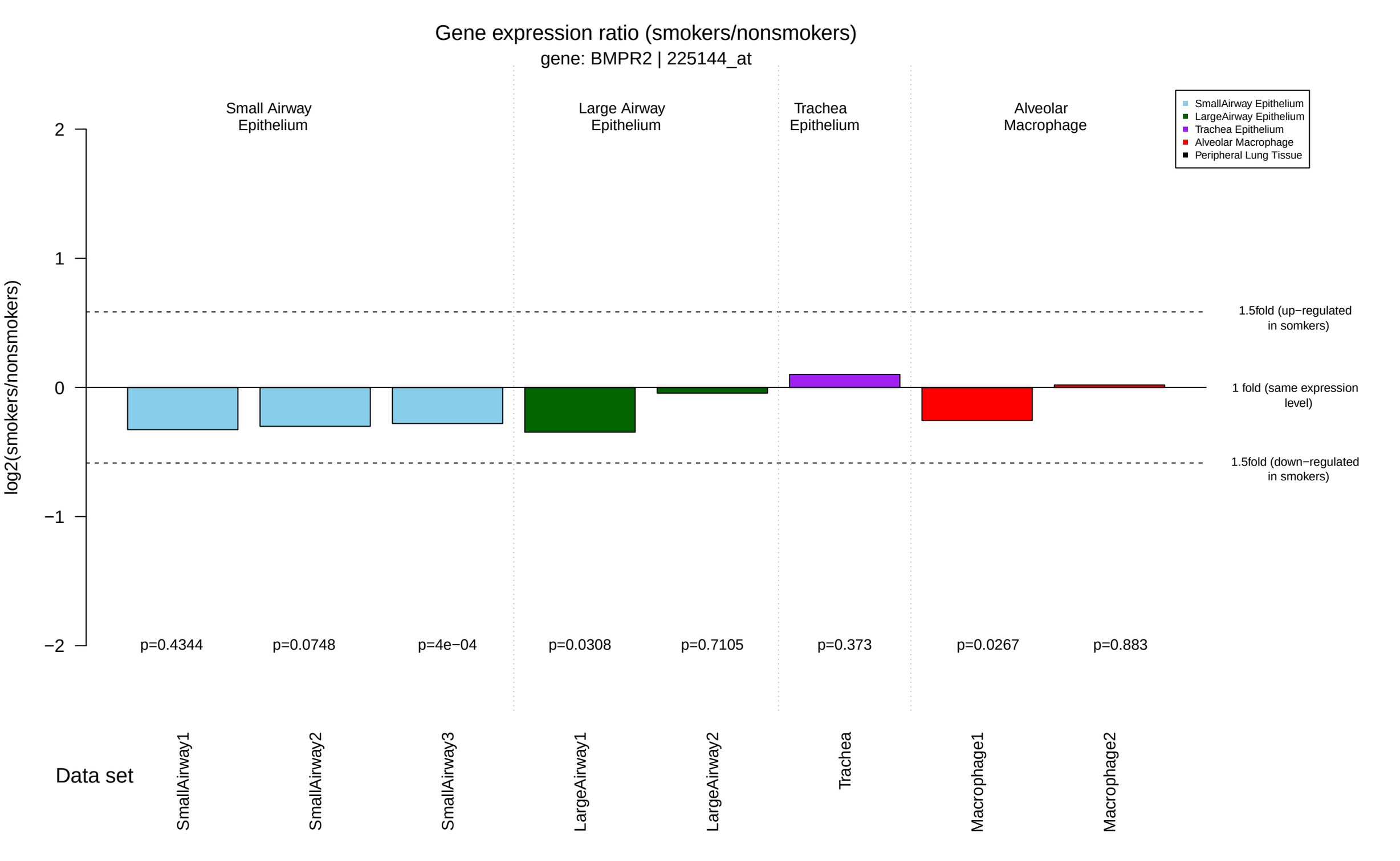
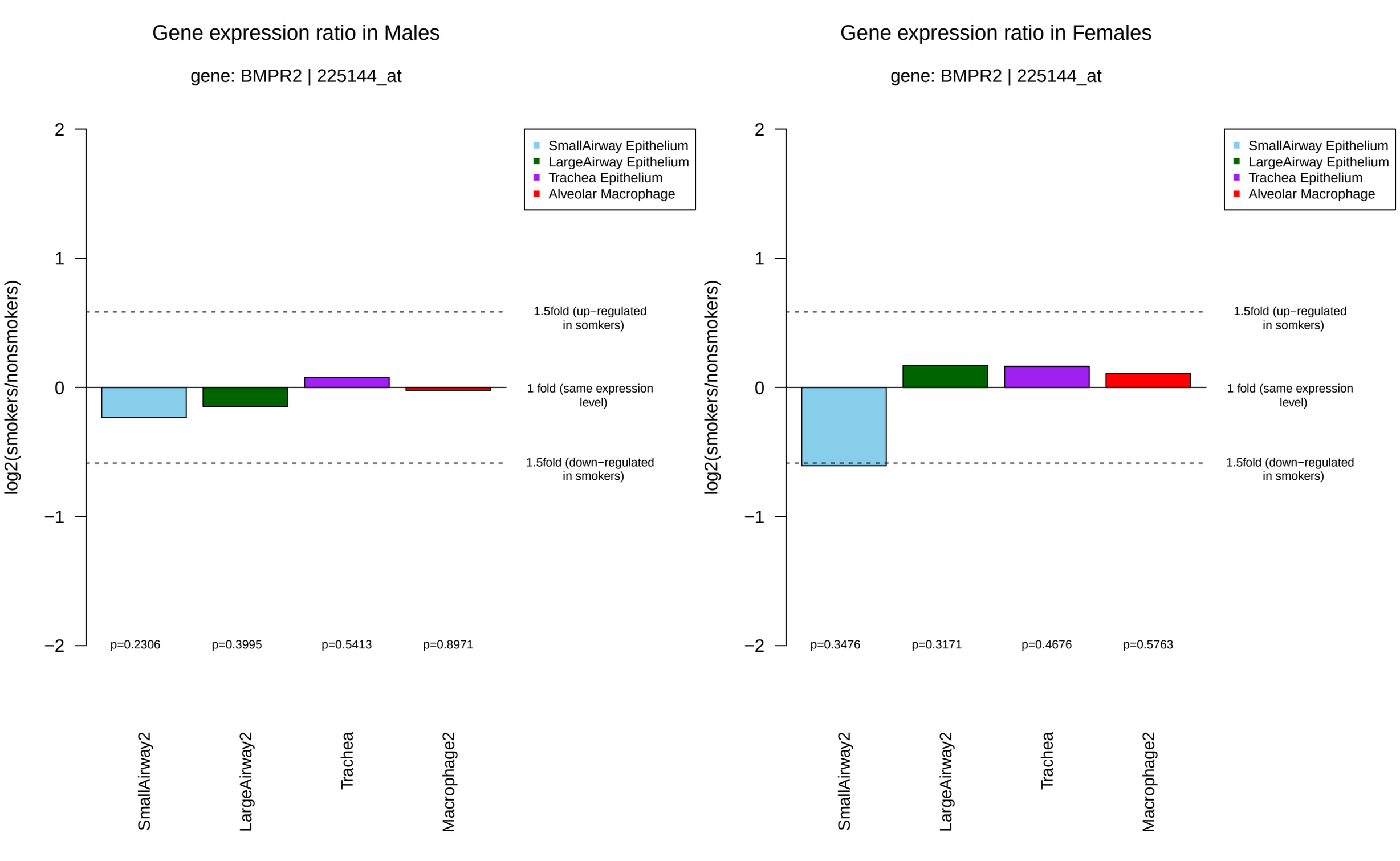
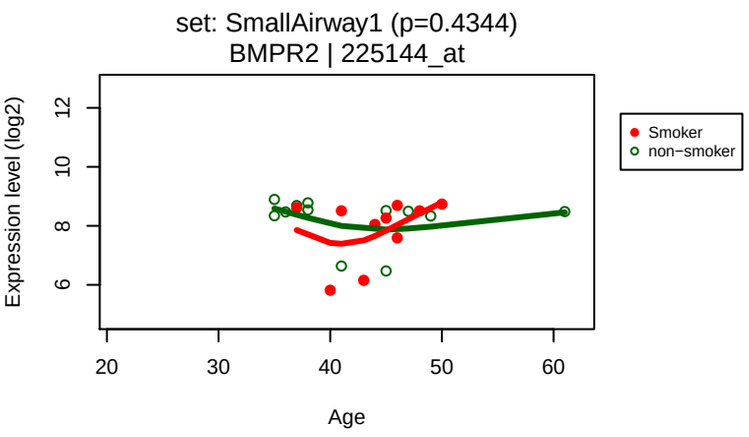
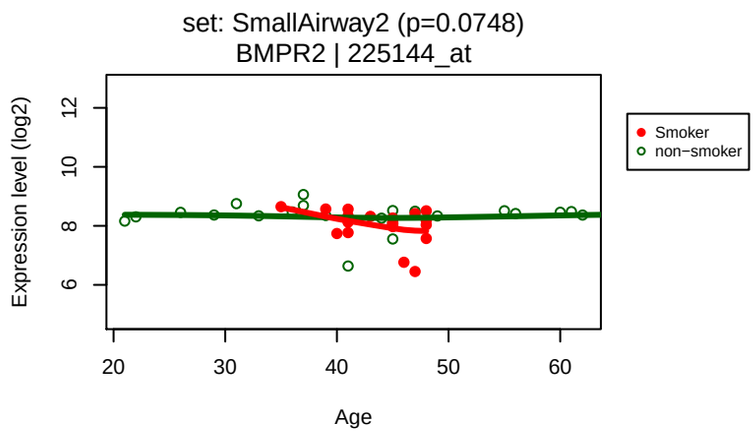
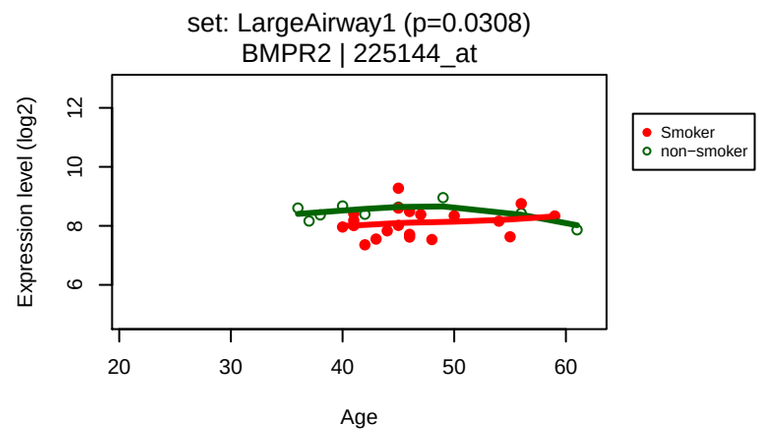

Gene expression of smokers and nonsmokers
gene: BMPR2 | 225144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4344
p=0.0748
p=4e−04
p=0.0308
p=0.7105
p=0.373
p=0.0267
p=0.883
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

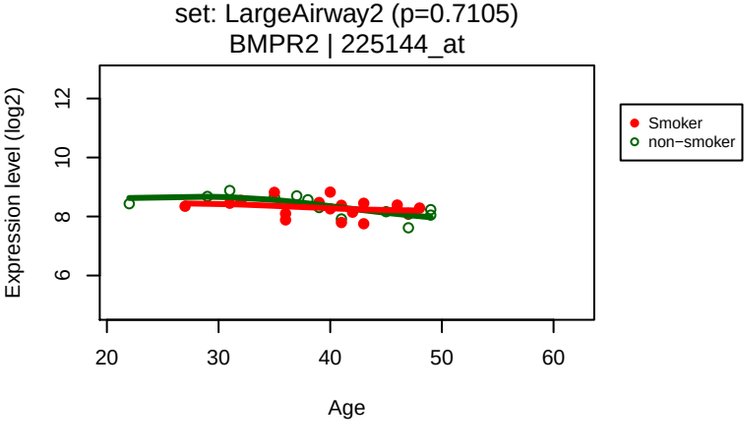

Gene expression ratio (smokers/nonsmokers)
gene: BMPR2 | 225144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4344
p=0.0748
p=4e−04
p=0.0308
p=0.7105
p=0.373
p=0.0267
p=0.883
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

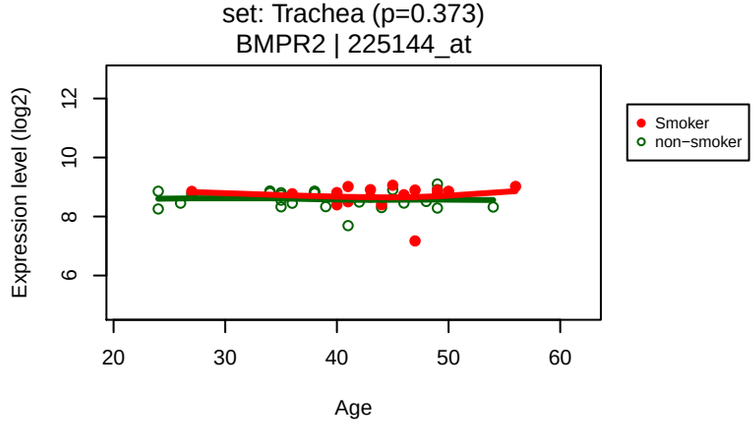

Gene expression ratio in Males
gene: BMPR2 | 225144_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2306
p=0.3995
p=0.5413
p=0.8971
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMPR2 | 225144_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3476
p=0.3171
p=0.4676
p=0.5763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

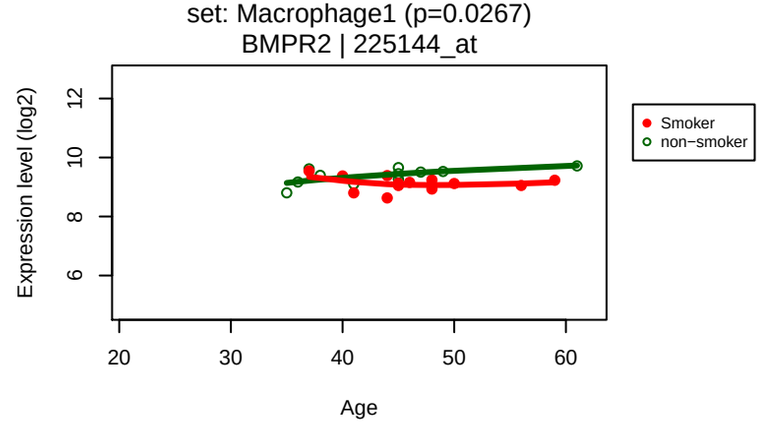

set: SmallAirway1 (p=0.4344)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker

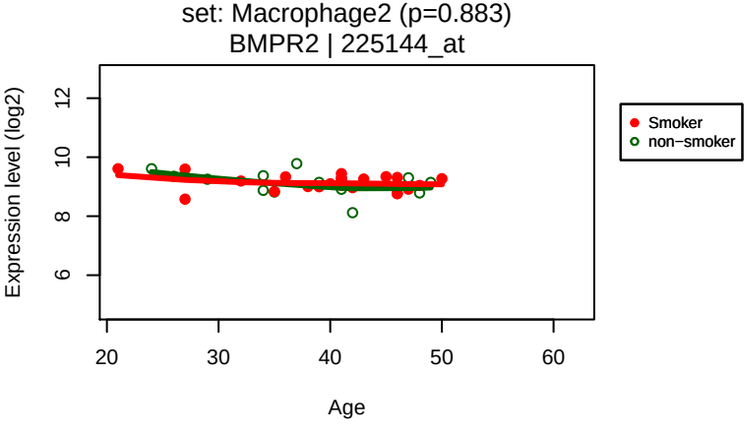

set: SmallAirway2 (p=0.0748)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker

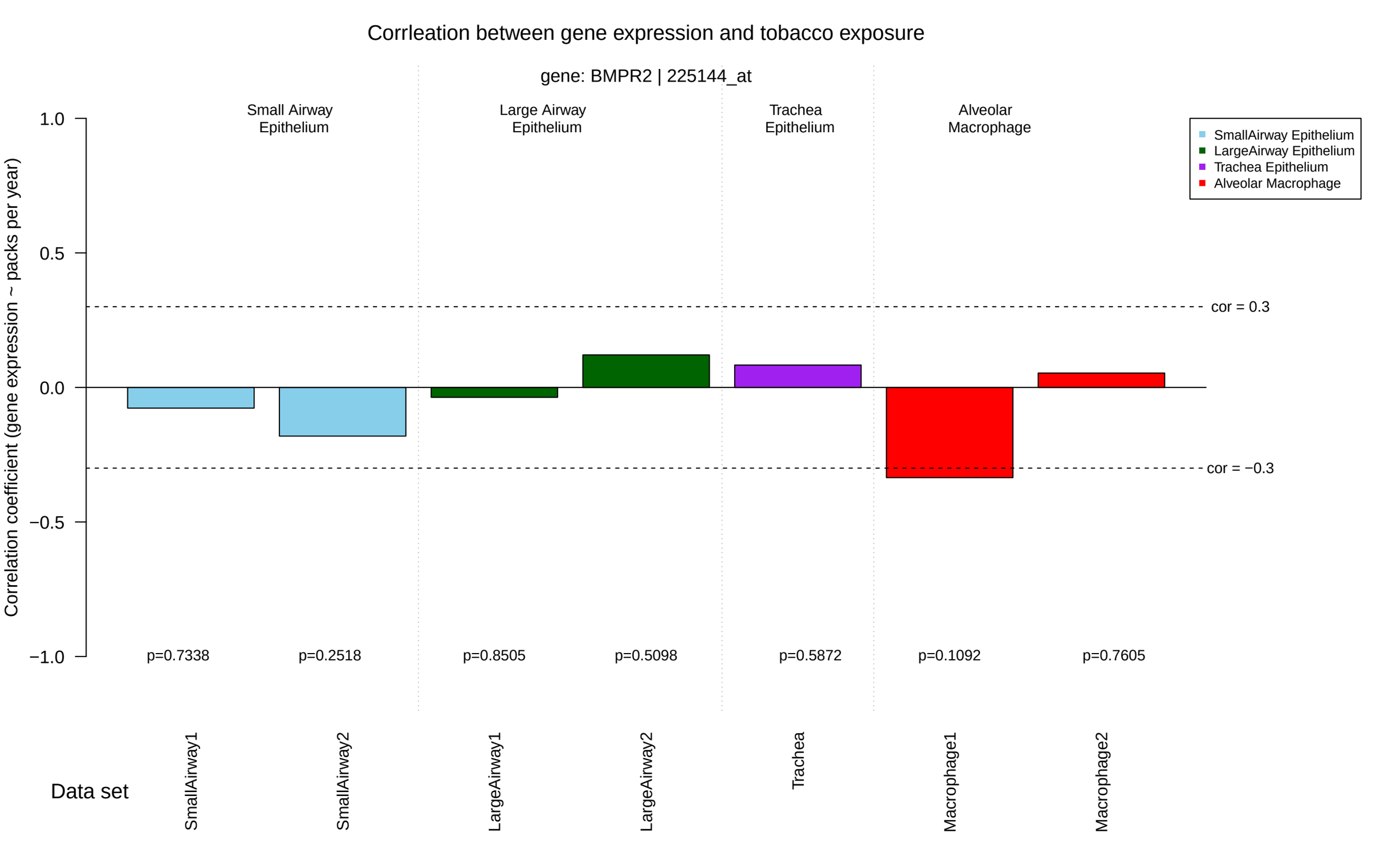

set: LargeAirway1 (p=0.0308)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7105)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.373)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0267)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.883)
BMPR2 | 225144_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMPR2 | 225144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7338
p=0.2518
p=0.8505
p=0.5098
p=0.5872
p=0.1092
p=0.7605
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2